

Theoretical Concepts

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Introduction

This chapter presents various food security indicators that can be derived from food data collected in national household surveys (NHS). It also introduces procedures to estimate the indicators as well as to standardize food consumption and expenditures data into dietary or monetary values.¹ Some of these procedures are done manually during the preparation of the datasets before executing ADePT-FSM, and others are automatically implemented in the software.

Food security indicators range from the prevalence of undernourishment to average consumption of various nutrients by source of food acquisition. These indicators are produced for different analytical groups based on the household and household's head characteristics collected in the NHS.

Section 1 of this chapter presents the food data collected in NHS, the procedures of standardization are further explained in section 2, and finally, indicators on food security and their related methodologies are introduced in the last section. Some practical examples related to the procedures of aggregation and standardization are provided in the annexes.

Food Data Collected in Household Surveys

Food data collected in household surveys are not standardized and strongly depend on survey design.² The way surveys are processed depends therefore on the unique characteristics of the food data collected. Specifically, this includes the sources of food acquisition, units of measurement of quantities

collected, and whether food data were collected in quantity and/or monetary values.

Sources of Food Consumption

Household food consumption ideally refers to the habitual consumption of food commodities, including nonalcoholic and alcoholic beverages.³ However, only a few surveys (such as yearly panel surveys) collecting information on food partakers are designed to capture household habitual food consumption. Usually the collection of food data in NHS refers to food acquired or consumed in or outside the sampled household during a given reference period.

Most food items acquired by the households are intended to be consumed by household members. Exceptions exist when acquired food is given to employees, guests, relatives, or pets. It may also be used to feed livestock, for small food businesses, or for resale. Therefore, to estimate the habitual household food consumption, the food not consumed by household members should be excluded in the food security analysis through proper identification at the collection stage. Ideally, any food losses and waste produced by the household should also be collected.

Households acquire food from various sources. Food can be purchased in markets, shops, food courts, restaurants, work canteens, from hawkers, etc. Food can also come from own production (farming, fishing, gathering, or hunting), or it can be withdrawn from private or business-owned stocks or received as payment or free from friends, relatives, and charity institutions.

During the data collection reference period, households may consume food items withdrawn from their own stocks, make bulk purchases, or accumulate stocks from their own production. For these reasons, especially for acquisition surveys, information on levels of initial and ending stocks should be properly reported during the food data collection period to avoid under- or overestimation of the habitual food consumption at the household level.

Summarizing, households usually consume food that is acquired from the following main sources:

- *Purchased food.* Food bought to be consumed inside the household, or food bought and consumed away from home, such as in restaurants, food courts, canteens, or from street vendors.

- *Nonpurchased food.* From own production (backyard gardens or farms), received free as gifts, donations, or transfers (including long-term food loan), received as payment in kind (including prepared food at workplaces), received as institutional food aid, or other (fishing, hunting, gathering, etc.).
- *Food stocks.* Composed of purchased or nonpurchased food items acquired prior to the starting date or during the reference period of food data collection.

Purchased Food

Purchased food items involve a payment by either cash or credit. Food purchases may be on a daily, weekly, or monthly basis depending on the type of food and payment of wages. Food can be purchased for consumption either within or outside the household. As the type of information (food quantities or monetary values) and the approach used to estimate dietary energy are different depending on whether the food is consumed inside or outside the home, it is important that the NHS captures food purchased for consumption at home and away from home separately.

In-House Consumption Perishable food commodities such as bread, milk, fresh fruits, and vegetables are usually purchased at shorter intervals (daily or weekly), while nonperishable commodities such as rice, flour, and sugar are usually acquired for consumption over a longer period of time (weekly or monthly). The payment of wages may be daily, weekly, fortnightly, monthly, or sometimes in relation to crop harvests (households usually purchase bulk quantities of specific food items in relation to the harvest cycles).

Consumed Away from Home Food purchases include food consumed away from home, such as drinks and ready-prepared meals from vendors, restaurants, food courts, school, work canteens, etc.

Nonpurchased Food

Own Consumption Households acquire some food commodities such as cereals, roots, tubers, vegetables, fruits, milk, and meat from their own production (from backyard gardens or farms). Some households consume all their food production, while others consume only part of it, selling the rest

for income. This type of food acquisition is commonly referred to as *own production* or *own consumption* or *self-production* and does not involve any monetary transactions. However, it is important to have proper estimates of the food quantity and monetary values acquired from this source. Own production may constitute an important source of food for particular household groups involved in agricultural livelihoods, especially in rural areas.

In Kind Households may also acquire and consume food items obtained free of charge, such as gifts, donations, or transfers from relatives and friends. In some countries, fishing, hunting and/or gathering provide a substantial amount of food to certain groups of the population. Various international or national institutions give some basic and essential food items to individuals or households as food aid on a regular or ad-hoc basis. Household members may also receive food from employers as part of payment (income in kind), especially those working in food activities such as vegetable cultivating, farming, or livestock food processing, or those working as food vendors. Food acquired from these sources may constitute an important part of the total household food consumption.

Stocks

Food stocks are usually comprised of nonperishable food such as cereals and preserved food. However, in some developed countries, it is also common that people stock perishable food by freezing it. Stocks are mainly accumulated by people in rural areas from own production during the harvest period or by urban rich households that can acquire bulk quantities at lower prices.

Food Consumption Data Collected in Quantities and/or Monetary Values

National household surveys collect data on food acquisition or consumption from purchases in monetary and quantitative terms. The data are collected at the food commodity level. Food purchased for consumption inside of the home refers to food items available in the market and usually expressed in well-specified standard units of weight or volume. Therefore, it is expected that the data collected in the survey have details related to quantity, unit of measurement, and cost (in monetary value). However, the information collected may vary depending on the source of food acquisition.

In some surveys, food purchased is collected only in monetary value. When quantities are not available, they need to be estimated using market retail prices. These prices correspond to local or regional markets or local food shops, or are derived from surveyed households for the reference period. It is recommended that those estimates are worked out at the data collection stage by the field interviewer in collaboration with the respondent.

Food acquired for household members' consumption from sources such as own production, gifts, and aid does not involve monetary value transactions. In this case the monetary values could be estimated using market retail prices or from surveyed households in the region. In most recent surveys, households are asked to report the monetary value of food from these sources as if the item were bought at the market.

Finally, household members may purchase and consume food and drinks outside the home. The type of food can vary from a well-defined "takeaway" commodity such as beer, carbonated beverage, hamburgers, corn on the cob, roasted chicken, and fried rice to a more general, "ordered" description such as dinner, meal, or breakfast. While the takeaway food is purchased in standard local units such as a portion or plate, the ordered meal is consumed in bars, restaurants, or work canteens, and it mixes food items according to a recipe that may differ among the food outlets. For takeaway food, it is possible to have information on food quantities in standard units along with monetary values; however, for ordered food, usually only monetary values are available. For these reasons, nutrients and calories of these two food groups are estimated following different procedures.

Table 2.1 summarizes the most common availability of data by source of food acquisition and the limitations that may appear when processing the data.

Unit of Measurement

On the one hand, the unit of measurement of quantities acquired can be standard, such as gram, kilogram, liter, or milliliter, or a local unit, such as bag, basket, cup, or heap. On the other hand, all factors to convert quantities into nutrient values are expressed in terms of nutrient content per 100 grams of the food product. To ensure a proper conversion of food quantities into nutrient values, it is important to have factors to convert local units into standard ones. For instance, if a household declared the acquisition of a heap of parsley, the interviewer has to inquire how many standard

Table 2.1: Most Common Availability of Data by Source of Food Acquisition and Possible Limitations in Processing Data

<i>Food source</i>		<i>Quantity details</i>	<i>Value details</i>	<i>Limitations</i>
Purchased for inside household food consumption.		yes	yes	Availability of grams or milliliters conversion factors related to the local quantity units of measurement.
Food purchased and consumed away from home.	(1) Prepared standard takeaway meals acquired outside home.	yes (Standard portions collected from food providers)	yes	Availability of grams or milliliters conversion factors related to the standard portions. The calorie and nutrient densities of the food products should be available in food composition tables.
	(2) Prepared meals acquired and eaten away from home (Ordered meals consumed in bars, restaurants, hotels, schools, workplaces, etc.).	no	yes	Frequently only food monetary values are available.
Food from own production		yes	yes/no	Retail prices have to be estimated at the local market or obtained from surveyed households in the region.
Food received free				
Food received as payment				
Food from stock				
Food obtained from institutional aid				

units (for instance grams) of parsley are usually in a heap in this specific region. In some countries, the unit of the National Statistical Office that is in charge of collecting prices also records gram weight equivalences of the local units of measurement.

Standardization Procedures

Indicators are expressed in terms of quantities, dietary energy, and monetary values per person per day. This means that all data collected in the survey and needed for the food security analysis should be standardized before being aggregated over time and space. Procedures of standardization start with the conversion of food monetary values or quantities collected in the NHS into dietary energy. These procedures are complex and strongly rely on the quality of the food data collected, the food composition data, and the quality of the food matching.⁴

Estimation of Dietary Energy

The human body requires energy for different purposes, including metabolic process, muscular activity, growth, and synthesis of new tissues. Humans

can access the required energy through the intake of energy-yielding macro-nutrients from foods that are protein, fats, carbohydrates (including fibers), and alcohol. Each contributes to the total calories but in a different proportion. Food energy is usually calculated on the macronutrients' content of the food product to which energy conversion factors are applied. In this way grams of nutrients are transformed into energy. There are two units for energy: calories (expressed in kcal) en joules (expressed in kJ). See table 2.2.

Joules is the recommended unit for energy due to historical reasons. However, the authors prefer to use calories as the unit of measurement for energy. Polyols and organic acids usually play a minor role and will therefore be omitted here.

The total amount of calories using the Atwater formula is calculated in ADePT-FSM as follows:

$$\text{Calories(Kcal)} = \text{Protein(g)} * 4 + \text{Fats(g)} * 9 + \text{Av. Carbohydrates(g)} * 4 \\ + \text{Fiber(g)} * 2 + \text{Alcohol(g)} * 7$$

Note: In the above equation: *Available carbohydrates = total carbohydrates – fibers.*

Macro- and micronutrient consumption are estimated by multiplying food quantities (collected in the survey or calculated based on prices) by nutrient values from national or regional food composition tables (FCT) or databases (FCDB). These nutrient values are usually expressed as grams (g), milligrams (mg), or micrograms (µg) of nutrients per 100 grams edible portion (EP) on a fresh basis.

The food reported in the survey must be matched to food in FCT/FCDB. This can be done only if the consumed foods are expressed in grams EP.⁵

Table 2.2: Atwater System

	<i>kJ/g</i>	<i>Kcal/g</i>
Protein	17	4
Fat	37	9
Available/total carbohydrate	17	4
Available carbohydrate in monosaccharide equivalents	16	3.75
(Dietary) fiber ^a	8	2
Alcohol (i.e., ethanol)	29	7
Organic acids ^b	13	3
Polyols ^b	10	24

Source: FAO 2002.

a. In case only a total carbohydrate value is available, no energy is attributed to the fiber value.

b. There are also specific conversion factors for individual polyols and organic acids.

When food quantities are given in milliliter or liter, they need to be converted in grams EP using density values.⁶

Unfortunately, the process to convert food quantities into nutrient content is not straightforward, mainly due to some limitations in the available data:

- Food quantities expressed in local units of measurement without a conversion factor into standard units
- Nonedible portions (e.g., bones, seeds, peels, etc.) are included in the reported food quantities but their proportion is not known to convert to EP
- Undefined food items such as dinner, lunch, meal, etc.
- No national or regional FCT/FCDB available
- The impossibility of getting nutrient values for local food items or food with a broad definition such as other cereals, other meat, etc.

Because of these limitations, the way to estimate the calories and nutrients consumed should be split into two procedures:

- *Procedure 1.* Used when food quantities can be expressed in grams EP and nutrient values of quantities are available for the food item
- *Procedure 2.* Used when food quantities do not exist or they cannot be converted to grams EP, but food expenditures are available

As households consume/acquire food for inside and outside household consumption, the estimated energy consumed/acquired is calculated using a combination of the two procedures. Table 2.3 shows the various cases where procedure 1 or 2 should be applied.

The steps to follow in both procedures are described below. A numeric example built on 19 households belonging to Region = 1, Area = Urban, and Income quintile = 2 is presented in annexes 2B and 2C.

Procedure 1: Estimation of Nutrients and Calories from Food Quantities

As was mentioned before, this procedure applies *only* when food quantities can be expressed in standard units (grams EP) and nutrient values are available.

Table 2.3: Data Availability

<i>Food quantity</i>	<i>Conversion factors to convert local unit of measurement into standard unit (grams or milliliters)</i>	<i>Food monetary value</i>	<i>Calories and nutrients conversion factors from FCT</i>	<i>Procedure to follow in the estimation of calorie and nutrients consumption</i>
YES	YES	YES	YES	PROCEDURE 1
YES	YES	YES	NO	PROCEDURE 2
YES	NO	YES	YES	PROCEDURE 1: If the quantities can be expressed in a standard unit using the food commodity price and the monetary value. PROCEDURE 2: If it is not possible to apply the
YES	NO	YES	NO	PROCEDURE 1
NO	YES/NO	YES	YES	PROCEDURE 2 PROCEDURE 1: If the quantities can be expressed in a standard unit using the food commodity price and the monetary value. PROCEDURE 2: If it is not possible to apply the
NO	NO	YES	NO	PROCEDURE 1 PROCEDURE 2

Six main steps are involved: the first two should be done manually, while the remaining are implemented in ADePT-FSM:

1. Standardization of the food quantities into grams or milliliters equivalent
2. Conversion of milliliters into grams
3. Adjustment of food quantities for nonedible portions
4. Estimation of grams of nutrients per household
5. Estimation of calories provided by each nutrient
6. Estimation of total calories per household

Step 1: Standardization of the Food Quantities into Grams or Milliliters Equivalent Food composition tables always refer to nutrient values per 100 grams edible food product. Therefore, the first important step is to ensure that all food quantities are converted into grams. When food quantities are expressed in a unit of measurement such as kilogram, gram, milliliter, liter, a can of 200 grams, or a bottle of 750 milliliters, conversion into gram or milliliter is straightforward. In these cases food quantities can be converted into grams or milliliter just multiplying the amount of quantity per 1,000, 1, 1, 1,000, 200, and 750, respectively. However, food quantities may be expressed in local units such as bag, basket, cup, glass, heap, plate, tin,

unit, etc. These local units are commonly used in many countries, and their gram equivalent differs by food product and country (in some cases, also by region within a country). If the local unit of measurement cannot be converted into standard units the analysis cannot be conducted; it is therefore important to estimate *the gram equivalent for the local units* by food item at the community level, based on a sample of food items. It is always recommended that households report the quantities of food consumed/acquired in a standard unit of measurement. When not possible the enumerator should convert local units into standard ones in the field at the time of data collection.

If after the data collection cases remain where food quantities are missing or cannot be expressed in grams, an indirect method can be applied. It has to be kept in mind however that this estimation is lowering the quality of the estimated energy and other nutrients, which should be taken into account in the interpretation of the results. This method estimates food quantities using food unit values (\$/gram) at the most representative level. As unit values change across time, the monetary values should be adjusted before computing the unit values. These are the steps:

1. Estimate deflators as the food price index of the month in which the household was interviewed divided by the average of the food price index over the survey period.
2. Deflate all food expenditures using the correspondent deflator.
3. Convert all food quantities into the same standard unit.
4. For each household compute the unit value per food item using the quantities in standard units and the associated deflated expenditures.
5. Compute the median unit values grouping households by region, area, and income quintile.⁷
6. Estimate the food quantities using the correspondent expenditure and the median unit value associated with the households.

Step 2: Conversion of Milliliters into Grams Nutrient values in FCT are usually expressed per 100 grams EP, so the food quantities expressed in milliliters should be converted into grams. To do so, it is necessary to have information on the food product's density coefficient (food product mass per unit of volume). Water at 4 degrees Celsius has a density of 1, whereas for the rest of the food items, the value can be less than 1 (e.g., oils) or more

than 1 (e.g., milk). Using density values,⁸ quantities in grams are estimated as below:

$$Food\ Quantity_{jh}(g) = Food\ Quantity_{jh}(ml) * density_j \left(\frac{g}{ml} \right)$$

where j stands for food product for which a valid density coefficient exists and h stands for household.

An example on how to transform food quantities expressed in units of volume into standard units is presented in annex 2A.

Step 3: Adjustment of Food Quantities for Nonedible Portions While food quantities acquired include nonedible portions such as peels, bones, seeds, etc., nutrient values in the FCT are usually expressed per 100 grams EP. For this reason, there is the need to transform “as purchased” quantities into edible ones.⁹ This transformation is done for each food commodity by applying the appropriate refuse factor.¹⁰ Some food commodities, such as rice, milk, or sugar, are 100 percent edible, but this is not the case for other food items such as bananas, meat with bones, peaches, and walnuts in a shell. Therefore, the refuse factor depends on the food product, and when it is expressed as a percentage varies from 0 (all edible) to 76 (walnuts in a shell) or to even more for some food items. In many FCT/FCDB, the refuse factor is not reported, but instead the edible coefficient (ranging between 0 and 1), which is defined as the edible portion of the food, i.e., the portion of the food without refuse. For example, an edible coefficient of 0.70 means that 70 percent of the food is edible while 30 percent is inedible.

Summarizing the food quantities obtained in grams after applying steps 1 and 2 are adjusted for nonedible portions with the formula:

$$Edible\ Food\ Quantity_{jh} = Food\ Quantity_{jh} * \left(1 - \frac{Refuse_j}{100} \right)$$

Or

$$Edible\ Food\ Quantity_{jh} = Food\ Quantity_{jh} * (Edible\ Coefficient_j)$$

where $Food\ Quantity_{jh}$ refers to the grams of food product j consumed/acquired by household h ; including the refuse (nonedible portion); *Edible*

$Food\ Quantity_{jh}$ refers to the edible grams (excluding bones, peels, seeds, etc.) of the food product j consumed/acquired by household h ; $Refuse_j$ refers to the percentage of nonedible grams in the food product j ; and $Edible\ Coefficient_j$ refers to the proportion of the food item being eligible.

Step 4: Estimation of Grams of Nutrients per Household The estimation of the nutrient content by food product is done by applying the nutrient values of the macronutrients (fat, protein, carbohydrates, fiber, and alcohol) as published in FCT/FCDB to the edible quantities expressed in grams. The total grams of nutrient, at household level, is the sum of the nutrients provided by each food item:

$$Q_{ih} = \sum_{j=1}^g ((Edible\ Food\ Quantity_{jh} * Nutrient\ Value_{ij})/100)$$

where i represents the nutrient (fat, available carbohydrate, protein, alcohol, or fiber) in household h ; j represents the food items for which there are valid nutrient values, and the food quantities are expressed in edible grams following steps 1, 2, and 3; and g is the total number of food items for which valid nutrient factors exist. The value of g is lower (or equal) to the total number of food items consumed/acquired by household h . Q_{ih} refers to the total grams of nutrient i in household h , and $Nutrient\ Value_{ij}$ is the total grams of nutrient i per 100 grams edible portion of the food product j .

Step 5: Estimation of Calories Provided by Each Nutrient The macronutrient consumption contributes to the estimated energy available for the human body. The amount of calories provided by protein, fats, available carbohydrates, fiber, and alcohol can be estimated using the Atwater system coefficients (see table 2.2 above):

$$N_{ih} = A_i * Q_{ih}$$

where N_i stands for dietary energy (kcal) provided by nutrient i in household h ; A_i represents the Atwater coefficient (kcal/gram) associated with the nutrient i ; and Q_{ih} refers to the total grams of nutrient i consumed in household h .

Step 6: Estimation of Total Calories per Household The total estimated energy from these g food items consumed/acquired per household is finally

derived summing from all foods the calories provided by protein, fats, available carbohydrates, fiber, and alcohol.

$$DEC_h = \sum_i N_{ih}$$

where DEC_h represents estimated energy (kcal) acquired/consumed by household h from the g food items for which a valid nutrient factor exists; and N_{ih} represents the dietary energy (kcal) provided by nutrient i in household h from the g food items for which a valid nutrient factor exists.

A numerical example following steps 3 to 6 of procedure 1 is presented in annex 2B.

Procedure 2: Estimation of Nutrients and Calories from Food Expenditure

Procedure 2 applies when (1) food quantities are not available, (2) the unit of measurement cannot be expressed in standard units, or (3) nutrient values for the food product are not available. For instance, this procedure is usually applied to food consumed away from home, prepared food, or food not well defined. Since procedure 2 generates estimations of lower quality, it is best to explore other possibilities to obtain the necessary data. For example, if nutrient values are not found in the national or regional FCT/FCDB, other FCT/FCDB should be consulted, or they should be calculated using recipes, etc.

Procedure 2 involves monetary values and follows four main steps, implemented in ADePT-FSM, to estimate the following:

- Proportion of calories provided by protein and fats
- Total missing calories
- Corresponding calories from protein, fats, carbohydrates (including fiber), and alcohol
- Missing grams of protein, fats, carbohydrates (including fiber), and alcohol

The steps are described below, and annex 2C provides a numerical example based on 20 households.

Step 1: Estimation of the Proportion of Calories Provided by Protein and Fats When procedure 1 was applied in a household to one or more food

items, the proportion of calories provided by these products from protein and fats are estimated with the formulas:

$$SPORT_h = \frac{\sum_{j=1}^g PROT_{jh}}{\sum_{j=1}^g DEC_{jh}} \quad \text{and} \quad SFAT_h = \frac{\sum_{j=1}^g FAT_{jh}}{\sum_{j=1}^g DEC_{jh}}$$

where for household h , $SPORT_h$ and $SFAT_h$ stand, respectively, for the proportion of dietary energy (kcal) derived from the content of protein and fat in the g food items to which procedure 1 was applied; for household h , $PROT_{jh}$ and FAT_{jh} calories stand for the dietary energy (kcal) derived from the content of protein and fat in the food item j ; and for household h , DEC_{jh} stands for total dietary energy (kcal) provided by the food item j .

Step 2: Estimation of Total Missing Calories Missing calories are those corresponding to food items for which it is not possible to apply procedure 1. The estimation of these calories is done using the median household calorie unit value (\$/kcal) at region/area/income quintile level. This median calorie unit value is calculated with data corresponding to the food items for which procedure 1 was applied, and is equal to the ratio between the expenditures (adjusted for temporal price fluctuations) and the corresponding dietary energy values.

$$UVal_h = \frac{\sum_{j=1}^g FDEXP_{jh}}{DEC_h}$$

where $UVal_h$ represents the calorie unit value of household h in local currency per kcal; $FDEXP_{jh}$ represents the total food expenditures occurred by household h to acquire the g food items to which procedure 1 was applied; and DEC_h represents the total dietary energy of household h brought by the n food items.

The final calorie unit value corresponds to the median household calorie unit value at region/area/income quintile level. This median calorie unit value is applied at the household level to the food expenditure of all the k food items for which corresponding calories were not estimated using procedure 1.

$$DEC_{kh} = \frac{FDEXP_{kh}}{UVal}$$

where k represents all the food products acquired by household h and for which no quantity or nutrient value exists (e.g., food consumed away from home); the sum of g and k corresponds to the total number of food items consumed/acquired by the household h ; DEC_{kh} represents the dietary energy of household h from the k food items; $FDEXP_{kh}$ represents the expenditures occurred by household h to acquire the k food items; and $UVal$ represents the median calorie unit value.

Step 3: Estimation of the Corresponding Calories from Protein, Fats, Carbohydrates (Including Fiber), and Alcohol The amount of calories provided by protein and fats are calculated applying the proportion of calories provided by them (computed in procedure 2, step 1) to the estimated missing calories (computed in procedure 2, step 2). Using the same notations as introduced in step 1 and step 2 of procedure 2, it becomes:

$$PROT_{kh} = DEC_{kh} * SPORT_h \quad \text{and} \quad FAT_{kh} = DEC_{kh} * SFAT_h$$

The calories provided by total carbohydrates (including fiber) and alcohol are calculated as the difference between the total estimated missing calories (computed in procedure 2, step 2) and the sum of calories provided by protein and fats (computed in procedure 2, step 3):

$$CAR_{kh} = DEC_{kh} - (PROT_{kh} + FAT_{kh})$$

where $PROT_{kh}$, FAT_{kh} and CAR_{kh} represent, respectively, the calories from protein, fats, carbohydrates (including fiber), and alcohol provided by the k food items acquired by household h .

Step 4: Estimation of the Missing Grams of Protein, Fats, Carbohydrates (Including Fiber), and Alcohol To obtain the missing grams of protein, fats, and carbohydrates the respective Atwater coefficients are applied to the estimated missing calories provided by each of these nutrients (computed in procedure 2, step 3).

Quality Consideration

The quality of the calculated energy content of food products will heavily influence the quality of the energy consumed/acquired and therefore the

food security statistics. Therefore, when interpreting the results several issues need to be taken into consideration:

- Food consumed outside the household is not always well collected. Therefore the value of food consumed/acquired over the reference period may be wrongly estimated.
- Habitual food consumption may not be captured well. There could be shortcomings in the survey or questionnaire design (e.g., food consumption is not collected over the year; list of food products is not exhaustive enough to fully reflect habitual consumption of the household; food consumption may be collected through interview with long recall periods; use of many units of measurement when collecting quantities of food consumed/acquired).
- The quality of the food matching between reported foods and those of the FCT. On the one hand, the higher the percentage of exact matches means the higher the quality of the dietary energy estimation. On the other hand, the higher the proportion of foods which cannot be converted to grams EP means the lower the quality of the estimate of dietary energy consumed/acquired.
- The quality of the food composition data. The FCT used should be adequate for the country, of high quality, and complete. There should be no missing data because they would lead to an underestimation of the energy values. In addition, the higher the proportion of foods for which no specific nutrient values can be attributed means the lower the quality of the energy and nutrient estimations.
- Treatment of outliers (i.e., implausible under- and overconsumption). Procedures to treat for outliers are not included in this book.

Estimation of Food and Total Consumption Expenditures and Income

If the data are reliable, total income is calculated as the sum of income of all household members. Otherwise, it can be approximated by the total household expenditure, which is equal to the sum of total consumption expenditure plus nonconsumption expenditures.

Total consumption expenditure includes the following:

- Food
- Clothing and footwear

- Gross rent, fuel, and power
- Furniture
- Medical care and health expenses
- Transport and communications
- Recreation, entertainment, education, and cultural services
- Miscellaneous (personal care, package tours, etc.)

Nonconsumption expenditures include direct and indirect taxes, insurance premiums, charity donations, social security contributions, and remittances or gifts to other households.

Adjustment to Account for Temporal Variability of Prices

As already discussed, the period usually covered by NHS is one year to account for the seasonal variations of food consumption and income. In this way, households report the acquired food over different months within a year. The monetary value of food commodities may vary not only among regions within a country due to extra costs as part of the regional trade chain, but also over the survey period due to price fluctuations or economic factors. Variations in the monetary values because of the geographical distribution of households are not removed for food security analysis because they are indicative of price differentials on an item within the country. However, in the estimation of food expenditure, total consumption expenditure, and income, it is important to consider inflation and deflation.

If food expenditures, consumption expenditures, and income have not been deflated before executing the ADePT-FSM, it can be done within the program by adjusting monetary values using monthly deflators. The deflators are calculated based on monthly food and consumer price indexes (FPI and CPI, respectively) associated with each household according to the month and year in which the household was surveyed. The deflators used to adjust food expenditure values are obtained as the ratio of the monthly FPI and the survey midperiod FPI, which is estimated as the average of all the monthly FPI during the survey period. The deflators used to adjust total consumption expenditure and income are obtained as the ratio of the monthly CPI and the survey midperiod CPI, which is estimated as the average of all the monthly CPI during the survey period.

Annex 2D shows an example of the calculation of food and total price deflators.

Table 2.4: Summary Table on Procedures of Standardization in ADePT-FSM

Conversion into dietary energy	Procedure 1	Steps 1 to 2	Manual
		Steps 3 to 6	Automatic
	Procedure 2	Steps 1 to 4	Automatic
Calculation of total expenditures			Automatic
Temporal adjustment			Automatic
Conversion in per person			Automatic
National and subnational inference			Automatic

Conversion in per Person per Day

The indicators are standardized and expressed in terms of per person per day, to remove variations due to household size and time period of data collection. Most of the data from household surveys do not allow for an analysis of the intrahousehold distribution of food consumption and expenditure. Therefore, it is assumed that both of them are equally distributed among household members. Some surveys collect information on the number of people who participated in the meal (partakers). When data on partakers are available food consumption statistics (excluding food expenditure) are calculated using the number of food partakers instead of the household size.

Inference at National and Subnational Levels

The users of food security statistics are interested in having the information disaggregated by population groups within the country. Therefore, the statistics estimated for the sampled households are adjusted to infer statistics at national and subnational levels; to do so, population statistical weights are applied to the data. The population weight is the household weight multiplied by the number of household members. The household weight is corrected for nonresponses, and it is calculated as the inverse of the probability of the household to be selected multiplied by the expansion factor.

Table 2.4 summarizes all the procedures that are automatic in ADePT-FSM and those that need to be undertaken manually during the preparation of the datasets prior to executing the software.

Indicators on Food Security

This section presents the food security indicators generated by ADePT-FSM and their respective methods of estimation. As discussed in the section on standardization, the indicators produced by ADePT-FSM are standardized

in per person per day and are representative at the national and subnational levels according to the survey sampling design.

Food Insecurity Indicators Produced by ADePT-FSM

Groups of Analysis

The food security indicators are derived at national and subnational levels. The subnational levels are subsamples of households in terms of geographic, demographic, or socioeconomic factors. Statistics are provided not only for groups of population but also for food groups or food commodities. The statistics derived using a parametric approach, such as prevalence of under-nourishment, can be produced only for the population groups for which the survey sample is representative.

Category of Population Groups ADePT-FSM allows for the analysis of food security indicators derived by population groups. These population groups include households' geographical location and household heads' socioeconomic or demographic characteristics. Table 2.5 presents all the population groups that can be analyzed using ADePT-FSM.

Food Commodity Groups The food commodity groups are defined by the user. Table 2.6 is an example of the classification used by the Statistics

Table 2.5: Population Groups

National
Regional
Urban and rural areas
Quintile of income
Household size
Gender of the household's head
Age of the household's head
Economic activity of the household's head
Education level of the household's head
Occupation of the household's head
Population group 1 defined by the user (e.g., marital status of household's head)
Population group 2 defined by the user (e.g., type of access to drinkable water)
Population group 3 defined by the user (e.g., does or does not receive institutional food aid)
Population group 4 defined by the user (e.g., refugee or not)
Population group 5 defined by the user (e.g., ethnicity)

Table 2.6: FAO Food Commodity Groups' Classification to Process Household Surveys

1	Cereals and derived products
2	Roots and tubers, and derived products
3	Sugar crops and sweeteners and derived products
4	Pulses and derived products
5	Nuts and derived products
6	Oil-bearing crops and derived products
7	Vegetables and derived products
8	Fruits and derived products
9	Stimulant crops and derived products
10	Spices
11	Alcoholic beverages
12	Meat (including poultry and pork) and derived products
13	Eggs
14	Seafood and derived products
15	Milk, cheese, and derived products
16	Vegetable oils and fats
17	Animal oils and fats
18	Nonalcoholic beverages
19	Miscellaneous and prepared food

Division of the FAO, which reflects the main food groups used for the food balance sheets.

Food Commodity Items The food commodity items analyzed are those listed in the NHS and are well-defined, such as fresh milk; long-grained rice; tomatoes; boneless, frozen mutton, etc. The list and number of food commodities are country- and survey-specific.

Indicators Produced by ADePT-FSM

The indicators listed in table 2.7 are derived for each category of population group (LCU refers to local currency). Indicators produced only at national, urban/rural areas, or regional levels are indicated with an asterisk (*).

The indicators listed in table 2.8 are derived for each food commodity group.

The indicators listed in table 2.9 are derived for each food item of the NHS.

Indicators and Methods of Estimation

The various indicators and related methods of estimation are presented below.

Table 2.7: Food Security Statistics Produced for Each Category of Population Groups

<i>General</i>	Number of sampled households Average household size Estimated population
<i>Access to diet and quality of diet</i>	
<i>Dietary energy and macro-nutrients</i>	Average dietary energy consumption (kcal/person/day) Average protein consumption (g/person/day) Average carbohydrates consumption (g/person/day) Average fats consumption (g/person/day) Average availability of vitamin A, retinol, and beta-carotene (mcg RAE/person/day) Average availability of vitamins B1, B2, B6, and C, calcium (mg/person/day), and vitamin B12 (mcg/person/day)
<i>Micronutrients</i>	Average availability of animal, nonanimal, heme, and nonheme iron (mg/person/day) Ratio of vitamins A and B12 available to required (%) Ratio of vitamins A, B1, B2, B6, B12, and C, and calcium available to recommended safe intake (%)
<i>Amino acids</i>	Average availability of essential amino acids: isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, valine, histidine, cystine, and tyrosine (g/person/day)
<i>Quality of diet</i>	Share of dietary energy consumption from protein (%) Share of dietary energy consumption from carbohydrates (%) Share of dietary energy consumption from fats (%) Share of animal protein in total protein consumption (%)
<i>Economic access to food</i>	
<i>Monetary value</i>	Average food consumption (LCU/person/day) Average total consumption (LCU/person/day) Average income (LCU/person/day)
<i>Price</i>	Average dietary energy unit value (LCU/1,000 kcal)
<i>Sources of acquisition</i>	Share of purchased food in total food consumption (in dietary energy) (%) Share of own produced food in total food consumption (in dietary energy) (%) Share of food consumed away from home in total food consumption (in dietary energy) (%) Share of food from other sources in total food consumption (in dietary energy) (%) Share of purchased food in total food consumption (in monetary value) (%) Share of own produced food in total food consumption (in monetary value) (%) Share of food consumed away from home in total food consumption (in monetary value) (%) Share of food from other sources in total food consumption (in monetary value) (%) Share of food consumption in total income (%) (Engel ratio)
<i>Responsiveness of demand to income inequality</i>	Income demand elasticity of dietary energy consumption Income demand elasticity of food consumption in monetary value Income demand elasticity of Engel ratio Dispersion ratio of food consumption in dietary energy (80/20) Dispersion ratio of food consumption in monetary value (80/20) Dispersion ratio of total consumption expenditure (80/20)
<i>Food inadequacy</i>	Minimum and average dietary energy requirements (kcal/person/day) Prevalence of undernourishment (%) (*) Depth of food deficit (kcal/person/day) (*)

Dietary Energy and Macronutrient Consumption

The average daily calories consumed by a representative individual in a population group of analysis are estimated as follows:

$$\text{Calories per person per day} = \frac{\sum_{h=1}^H (hh_wgt_h * DEC_h)}{\sum_{h=1}^H (hh_size_h * hh_wgt_h * num_days_h)}$$

Analyzing Food Security Using Household Survey Data

Table 2.8: Food Security Statistics Produced for Each Food Commodity Group

<i>Access to diet and quality of diet</i>	
<i>Dietary energy and macronutrients</i>	Average dietary energy consumption (kcal/person/day)
	Average protein consumption (g/person/day)
	Average carbohydrates consumption (g/person/day)
	Average fats consumption (g/person/day)
<i>Quality of diet</i>	Contribution of food groups to total dietary energy consumption (%)
	Contribution of food groups to total protein consumption (%)
	Contribution of food groups to total carbohydrates consumption (%)
	Contribution of food groups to total fats consumption (%)
	Average protein consumption per kcal (g/1,000 kcal)
	Average carbohydrates consumption per kcal (g/1,000 kcal)
<i>Micronutrients</i>	Average fats consumption per kcal (g/1,000 kcal)
	Average availability of vitamin A, retinol, and beta-carotene (mcg RAE/person/day)
	Average availability of vitamins B1, B2, B6, and C, and calcium (mg/person/day), and vitamin B12 (mcg/person/day)
	Average availability of animal, nonanimal, heme, and nonheme iron (mg/person/day)
<i>Amino acids</i>	Contribution of food groups to micronutrient availability (%)
	Average availability of essential amino acids: isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, valine, histidine, cystine, and tyrosine (g/person/day)
	Contribution of food groups to amino acid availability (%)
<i>Economic access to food</i>	
<i>Monetary value</i>	Average food consumption (LCU/person/day)
<i>Price</i>	Average dietary energy unit value (LCU/1,000 kcal)
	Average protein unit value (LCU/100 g)
	Average carbohydrates unit value (LCU/100 g)
	Average fats unit value (LCU/100 g)

Table 2.9: Food Security Statistics Produced for Each Food Commodity

<i>Access to diet and quality of diet</i>	
<i>Dietary energy and macronutrients</i>	Average edible food quantity (g/person/day)
	Average dietary energy consumption (kcal/person/day)
	Average protein consumption (g/person/day)
<i>Micronutrients</i>	Average availability of vitamin A, retinol, and beta-carotene (mcg RAE/person/day)
	Average availability of vitamins B1, B2, B6, and C, and calcium (mg/person/day), and vitamin B12 (mcg/person/day)
	Average availability of animal, nonanimal, heme, and nonheme iron (mg/person/day)
<i>Amino acids</i>	Average availability of essential amino acids: isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, valine, histidine, cystine, and tyrosine (g/person/day)
<i>Economic access to food</i>	
<i>Monetary value</i>	Average food consumption (LCU/person/day)
<i>Price</i>	Dietary energy unit value (LCU/1000 kcal)

$$DEC_h = \sum_{j=1}^{ag} DEC_{hj} + \sum_{j=1}^k DEC_{hj}$$

The average daily macronutrients (protein, carbohydrates, and fat) consumed by a representative individual in a population group is estimated as follows:

$$\text{Macronutrients per person per day} = \frac{\sum_{h=1}^H (hh_wgt_h * Nutrient_h)}{\sum_{h=1}^H (hh_size_h * hh_wgt_h * num_days_h)}$$

$$Nutrient_h = \sum_{j=1}^g Nutrient_{hj} + \sum_{j=1}^k Nutrient_{hj}$$

$$Nutrient_{jh} = \left(\left(\frac{fq_{jh} * lg_j}{100} \right) * \left(1 - \frac{refuse_j}{100} \right) \right)$$

where H is the total number of sampled households belonging to the population group of analysis; hh_wgt_h is the household weight (expansion factor divided by the probability of the household to be sampled) of household h ; hh_size_h is the total number of members (household size) in household h ; num_days_h is the number of days of the food data reference period for household h ; DEC_{hj} refers to the calories consumed of food item j by household h ; g is the number of food items in household h , for which the nutrient content is estimated applying procedure 1; k is the number of food items in household h , for which the nutrient content is estimated applying procedure 2; $Nutrient_h$ is the total amount of macronutrients in household h ; $Nutrient_{hj}$ is the amount of macronutrients in food item j in household h ; fq_{jh} is the quantity of the food item j consumed/acquired by household h and expressed in grams “as purchased” (includes the nonedible part); lg_j refers to the grams of micronutrients per 100 grams edible portion of the food item j (as in FCT); and $refuse_j$ is the refuse factor (nonedible part) of the food item j expressed in percentage.

Micronutrient Availability

A proper intake of macronutrients in terms of a balanced diet is not enough for human beings to conduct a healthy life if they do not consume adequate amounts of minerals, vitamins (micronutrients), and indispensable amino acids. The micronutrients analyzed by the ADePT-Food Security Module are the A vitamins, ascorbic acid, thiamine, riboflavin, B6, cobalamin, and the minerals calcium and iron. The indispensable amino acids analyzed are isoleucine, leucine, lysine, threonine, tryptophan, valine, histidine, methionine and cystine, and phenylalanine and tyrosine.

Most NHS collect data on food acquisition rather than consumption. Further, the content of micronutrients in food may vary from the moment of its acquisition to its consumption because of several reasons including storage conditions and the way the food is processed. Moreover, the presence of other substances in the food may inhibit or enhance nutrient absorption.¹¹ Therefore, the derived estimates are indicative, and they should not be interpreted as a result of the evaluation of intake by individuals in the population groups. That is why the term *availability* is used instead of *consumption*.

In the micronutrients assessment, neither supplementation nor fortification is taken into consideration because such information usually is not collected in the surveys.

The equation applied to estimate the micronutrients available for consumption is given below:

$$\text{Micronutrients per person per day} = \frac{\sum_{h=1}^H (hh_wgt_h * Nutrient_h)}{\sum_{h=1}^H (hh_size_h * hh_wgt_h * num_days_h)}$$

$$Nutrient_h = \sum_{j=1}^n Nutrient_{hj}$$

$$Nutrient_j = \left(\left(\frac{fq_{jh} * lg_j}{100} \right) * \left(1 - \frac{refuse_j}{100} \right) \right)$$

where H is the total number of sampled households belonging to the population group of analysis; hh_wgt_h is the household weight (expansion factor divided by the probability of the household to be sampled) of household h ; hh_size_h is the total number of members (household size) in household h ; num_days_h is the number of days of the food data reference period for household h ; n is the total number of food items (excluding those consumed away from home¹²) in household h ; $Nutrient_h$ is the amount of micronutrients available in household h (excluding those consumed away from home); $Nutrient_{hj}$ is the amount of micronutrients available in food item j in household h ; fq_{jh} is the quantity of the food item j consumed/acquired by household h and expressed in grams “as purchased” (includes the nonedible part); lg_j refers to the grams of micronutrients in 100 edible grams of the food item j

(as in FCT); and $refuse_j$ is the refuse factor (nonedible part) of the food item j expressed in percentage.

For the micronutrients' ADePT-FSM analysis, it is worth clarifying some concepts on vitamin A and iron.

More than one unit of measurement can be found in the literature when referring to vitamin A. The ADePT-FSM was developed to express the availability of vitamin A in terms of retinol activity equivalent (RAE). The use of μg RAE rather than μg retinol equivalent (RE) or international units (IU) is preferred when calculating and reporting the amount of the total vitamin A in mixed foods or assessing the amount of dietary and supplemental vitamin A consumed (see National Academy of Sciences [NAS] 2001). According to NAS the conversion of retinol and pro-vitamin A carotenoids into vitamin A is as follows:

$$\begin{aligned} \text{Vitamin A in } \mu\text{g RAE} = & \mu\text{g retinol} + \frac{\mu\text{g } \beta\text{carotene}}{12} \\ & + \frac{\mu\text{g } \alpha\text{carotene or } \beta\text{cryptoxanthin}}{24} \end{aligned}$$

Regarding iron, it can be distinguished as animal or nonanimal. The first refers to meat, fish, eggs, milk, and cheese, and their derived products. Another type of iron classification is with respect to the mechanism of its absorption: heme and nonheme. The latter is present in food of both animal and nonanimal origin, whereas the former can be found only in meat and fish (as it is derived from hemoglobin and myoglobin).¹³

Amino Acids

Amino acids are the building blocks of proteins. Some of their functions are building cells, protecting the body from viruses or bacteria, repairing damaged tissue, providing nitrogen, and carrying oxygen throughout the body. They can be classified as dispensable or indispensable. The latter are also called essential amino acids (EAA) and cannot be synthesized by the human body. Therefore, they should be supplied to the body through the consumption of protein in food.¹⁴

The indispensable amino acids analyzed in ADePT-FSM are isoleucine, leucine, lysine, threonine, tryptophan, valine, histidine, methionine and

cystine, and phenylalanine and tyrosine. ADePT-FSM estimates the average daily per person grams of indispensable amino acids available for consumption and the equations used are:

$$\text{Amino acid per person per day} = \frac{\sum_{h=1}^H (hh_wgt_h * AA_h)}{\sum_{h=1}^H (hh_size_h * hh_wgt_h * num_days_h)}$$

$$AA_h = \sum_{j=1}^n AA_{jh}$$

$$AA_{jh} = \left(\left(\frac{fq_{jh} * lg_j}{100} \right) * \left(1 - \frac{ref}{100} \right) * \left(\frac{pd}{100} \right) \right)$$

where H is the total number of sampled households belonging to the population group of analysis; hh_wgt_h is the household weight (expansion factor divided by the probability of the household to be sampled) of household h ; hh_size_h is the total number of members (household size) in household h ; num_days_h is the number of days of the food data reference period for household h ; n is the total number of food items (excluding those consumed away from home¹⁵) in household h ; AA_h is the amount of amino acid available in household h (excluding those consumed away from home); AA_{jh} is the amount of amino acid available in food item j in household h ; fq_{jh} is the quantity of the food item j consumed/acquired by household h and expressed in grams “as purchased” (includes the nonedible part); lg_j refers to the grams of amino acid in 100 edible grams of the food item j (as in FCT); ref is the refuse factor (nonedible part) of the food item j expressed in percentage; and pd is the protein digestibility of the food item j expressed in percentage.

Balanced Diet

A balanced diet is a diet that provides energy and all essential nutrients for growth and a healthy and active life. Since few foods contain all the nutrients required to permit the normal growth, maintenance, and functioning of the human body, a variety of food is needed to cover a person’s macro- and micronutrient needs. Any combination of foods that provides the correct amount of dietary energy and all essential nutrients in optimal amounts and proportions is a balanced diet (CFS 2012).

A joint WHO/FAO expert group established guidelines for a balanced diet (WHO 2003). These guidelines are related to effects on the chronic nondeficiency diseases. According to expert opinion, a balanced diet exists when the following conditions are met:

- The proportion of dietary energy provided by protein is in the range of 10–15 percent.
- The proportion of dietary energy provided by fats is in the range of 15–30 percent.
- The proportion of total dietary energy available derived from carbohydrates is in the range of 55–75 percent.

From surveys collecting food consumption or acquisition, it is not possible to assess if a population group consumes a balanced diet, because there is no information about how people combine the food they consume or about intrahousehold differences in food consumption. However, from the data collected it is possible to infer whether or not households have access to a balanced diet.

Monetary Values

The average daily food expenditure of a representative individual in a population group is estimated as follows:

$$\text{Food monetary value per person per day} = \frac{\sum_{h=1}^H (hh_wgt_h * FMV_h)}{\sum_{h=1}^H (hh_size_h * hh_wgt_h * num_days_h)}$$

$$FMV_h = \sum_{j=1}^n FMV_{jh}$$

where H is the total number of sampled households belonging to the population group of analysis; hh_wgt_h is the household weight (expansion factor divided by the probability of the household to be sampled) of household h ; hh_size_h is the total number of members (household size) in household h ; num_days_h is the number of days of the food data reference period for household h ; n is the total number of food items (including those consumed away from home) in household h ; FMV_h is the food monetary

value of household h ; and FMV_{jh} is the food monetary value of food item j in household h .

A similar formula is applied to estimate the average daily total consumption expenditure and income of a representative individual in a population group.

Price

Food prices are important determinants of food security. ADePT-FSM calculates the calorie unit value of dietary energy expressed in monetary value per 1,000 kcal. At the population group level, the software computes the calorie unit value at household level and then estimates the correspondent mean. At food item and food group levels, because some food items or food groups may present only a few cases, the median calorie unit value is used instead of the mean. Finally, the calorie unit values do not include the cost of the energy required to cook food.

In addition to the monetary value of 1,000 kcal, macronutrient unit monetary values are also provided, and they are expressed in monetary terms per 100 grams of nutrients.

Responsiveness of Food Demand to Income

The income elasticity of the demand of food is measured through the responsiveness of dietary energy, food expenditure, or the Engel ratio to a variation in income. In other words, this relative responsiveness depicts the relationship between acquired food and income, described by the Engel curve. The estimates assume that substitution among food commodities occurs for different income levels and that *food commodity prices are constant*.

For a given country, it is assumed that household dietary energy consumption per person can be linked to household income per person by the following regression equation (FAO 1996):

$$x_h = \beta_0 + \beta_1 * \ln(V_h) + u_h$$

where β_0 and β_1 are parameters of the equation; x_h is the demand of food expressed in terms of dietary energy, food expenditure, or Engel ratio of household h on a per person basis; V_h is the income per person in household h ; and u_h is the random variation of food demand across households.

In ADePT-FSM the individual household data are grouped by income decile classes, and the average food consumption and income per person in each income decile class is inferred and used to estimate the parameters of the equation. Therefore the equation becomes:

$$\bar{x}_j = \alpha_0 + \alpha_1 * \ln(\bar{V}_j) + u_j$$

where α_0 and α_1 are parameters of the equation; and $\bar{x}_j$ is the inferred demand of food expressed in terms of dietary energy, food expenditure, or Engel ratio of the income class j on a per person basis.

$$\bar{x}_j = \frac{\sum_{h=1}^{H_j} (f_h * x_h)}{pop_j}$$

$\bar{V}_j$ is the inferred per person income of the income class j .

$$\bar{V}_j = \frac{\sum_{h=1}^{H_j} (f_h * V_h)}{pop_j}$$

H_h refers to the total number of sampled households in income decile j ; x_h is the demand of food expressed in terms of dietary energy, food expenditure, or Engel ratio of household h ; and pop_j is the inferred total population in the income decile j .

$$pop_j = \sum_{h=1}^{H_j} (hh_size * hh_wgt_h)$$

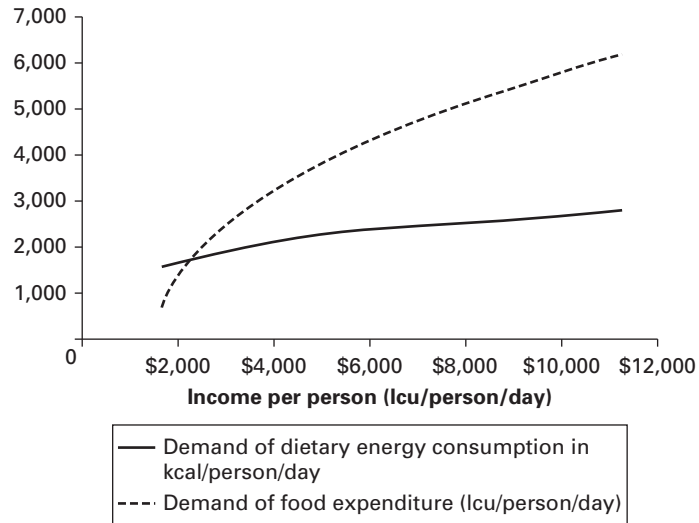
f_h is the inferred total number of people represented by household h .

$$f_h = (hh_size * hh_wgt_h)$$

hh_size_h is the total number of members (household size) in household h ; hh_wgt_h is the household weight (expansion factor divided by the probability of the household to be sampled) of household h ; and u_j is the random variation of the average food demand across income decile classes.

Figure 2.1 shows an example of the demand of food consumption as a function of income. This example is from real country data and represents the average values of food demand (in terms of dietary energy and food expenditure) by average values of income (estimated for income decile groups of the population).

Figure 2.1: Example of Food Consumption Demand as Function of Income



The elasticity of food consumption with respect to income is:

$$\eta = \left(\frac{\partial x}{\partial V} \right) * \frac{V}{\bar{x}} = \frac{\alpha_1}{V} * \frac{V}{\bar{x}} = \frac{\alpha_1}{\bar{x}}$$

Therefore, the income elasticity (η) of food demand (represented by the mean per person μ) can be estimated as:

$$\eta = \frac{\alpha_1}{\mu}$$

Reviewing: This expression of elasticity allows the estimation of the elasticity by income deciles.

$$\eta_j = \left(\frac{\hat{\alpha}_1}{\hat{x}_j} \right) = \left(\frac{\hat{\alpha}_1}{\hat{\alpha}_0 + \hat{\alpha}_1 * \ln(\bar{V}_j)} \right)$$

where $\hat{\alpha}_1$ is the estimated slope of the Engel function; and $\hat{x}_j$ corresponds to the estimated fitted mean of dietary energy, food expenditure, or Engel ratio of the j_{th} income decile.

Food Consumption Statistics by Sources of Acquisition

As discussed previously, households acquire food from different sources, including purchases, own production, aid, and as payment for labor. The ADePT-FSM analyzes four main sources of food acquisition: purchases (excluding food consumed away from home), own production, consumed away from home, and all other sources. The latter includes food received as aid, gift or payment for labor, hunting, and wild harvesting.

The contribution of each food source to total food consumption in both monetary and dietary energy terms varies depending on the population group of analysis. For instance, it is expected that the share of food consumption from own production in rural households is higher than that in urban households.

The share of food expenditure in total income is also called the Engel ratio. Low-income households spend a large percentage of their total consumption expenditure on food. With higher income, the food ratio declines following Engel's law, which states that the proportion of income spent on food decreases with increasing income, because food is a basic primary need.

Measures of Inequality

Dispersion Ratios The dispersion ratios measure inequality between the two extreme income quintile groups. They are calculated using as reference the average values corresponding to the first quintile. For instance, one ratio is defined as the average food consumption (in terms of dietary energy or monetary values) in the highest income quintile divided by the correspondent average food consumption in the lowest one.

Coefficient of Variation The coefficient of variation (CV) is a relative measure of inequality in a given distribution. In principle, a *direct* estimate of the variability in the distribution of dietary energy consumption (DEC) could be obtained through a measure of the empirical dispersion of individuals' consumption from a survey. There are, however, several reasons why this may be problematic. Household food consumption data (collected in surveys) on a per person basis are very likely to be more dispersed than the actual per person yearly average of food consumption in the population. This is because of the presence of "spurious" variability (introduced both

systematically through features of survey design and accidentally due to nonsampling errors) related to the following:

- *Survey rounds of data collections usually spread over the year.* This is done to avoid introducing biases in the estimation of mean consumption, when consumption of food is known to be varying over the seasons. Unfortunately, spreading data collection over the seasons means that seasonal *variability* in consumption (which should not be considered in estimating the variability of the average year consumption in the population) is still present.
- *Missing data and outliers.* For example, nonsampling errors that are associated with errors in recall, under- or overreporting, incompleteness of data collection forms, especially with reference to food consumed away from home, interview effects, etc.
- *Surveys collecting food acquisition instead of food consumption.* Food acquisition surveys may overestimate the distribution of calories across households because the variability *within* households will be confounded with the variability *between* households. Calories can be acquired through durable foods such as cereals to be stocked and consumed over a long period of time and not during the reference food collection period.
- *Expected variability.* During the year, there is an expected variability in adequately nourished households, for instance, the result of a party given by the households. This variability is considered as an excess variability, since we are interested in capturing the habitual food consumption.

All these factors might induce a systematic positive bias in the estimate of the *variability* parameter of the distribution. This bias is difficult to reduce once survey data have been collected. Cleaning the data to identify outliers and missing values can help reduce the potential bias, but this may introduce a certain degree of subjectivity in the analysis that should not go unnoticed. In addition, when the distribution in the population is skewed, as seen in two-stage sampling (commonly used in household income expenditure surveys), a systemic bias in the estimate of variability indicators can result.

All these considerations raise reservations about the possibility of obtaining a reliable estimate of DEC variability through the observed empirical variance of individual household data in a survey. Therefore,

the estimation of the CV of DEC can be derived as the combination of two sources of variability of DEC: one due to income and the other due to other factors. Indeed, while the role of income in explaining DEC and its variability within a population is at the heart of all theories of poverty and economic development, there are many other factors inducing variability in DEC. These factors have to be considered physiological in a population, and should therefore be tolerated.

The overall value of the CV is then obtained following the steps described below:

Step 1: Estimation of the CV of DEC Tabulated by Income Individual households are grouped by classes of income decile, and the average per person food consumption in each class of income is inferred. By averaging within an income class, most of the variations in the level of DEC because of factors that are not strongly correlated with income are clearly netted out. The resulting measure of CV should thus properly be interpreted as an estimate of the component of the total variability of DEC in the population tabulated by income, which we term $CV_{x/v}(x)$.

The coefficient of variation of DEC tabulated by income is defined as:

$$CV_{(x/v)} = \frac{\sigma_{(x/v)}}{\mu_x}$$

where $\sigma_{(x/v)}$ is the standard deviation of the distribution of the average per person dietary energy consumption of income decile groups and is derived from the formula:

$$\sigma_{(x/v)} = \sqrt{\frac{\left[\sum_{j=1}^{10} (f_j * x_j^2) - \frac{\left(\sum_{j=1}^{10} f_j * x_j \right)^2}{pop} \right]}{pop - 1}}$$

μ_x is the average per person dietary energy consumption at the income decile level and is derived from the formula:

$$\mu_x = \frac{\sum_{j=1}^{10} (f_j * \bar{x}_j)}{pop} = \frac{\sum_{h=1}^H (f_h * x_h)}{pop}$$

where j refers to income decile group; h refers to household; H is the total number of sampled households in the survey; Hj is the total number of

sampled households in income decile j ; $\bar{x}_j$ is the average per person dietary energy consumption of income decile j ; x_h is the average per person dietary energy consumption of household h ; and pop is the inferred total population.

$$pop = \sum_{h=1}^H (hh_size_h * hh_wgt_h)$$

f_j is the inference total number of people in income decile j .

$$f_i = \sum_{h=1}^{H_j} (hh_size * hh_wgt)_h$$

f_h is the inference total number of people represented by household h .

$$f_h = (hh_size * hh_wgt)_h$$

hh_size_h is the total number of members (household size) in household h ; and hh_wgt_h is the household weight (expansion factor divided by the probability of the household to be sampled) of household h .

Step 2: Estimation of CV of DEC Because of Other Factors If it is true that people tend to consume according to their respective dietary energy requirements (DER), and as long as there is an interindividual variation in DER, there will be variation in DEC due to physiological factors. For this reason, a component reflecting the variability of DEC induced by the factors determining the variability of DER, $CV_{x/r}(x) = CV(r)$ is also considered to estimate the total CV. This variation of dietary energy due to requirements is estimated taking into account the coefficient of variation of three components: body weight, physical activity level (PAL), and measurement error. The coefficient of variations due to body weight and PAL are estimated under the assumption of the lognormality. The regression equations used for estimating the basal metabolic rate (BMR) given a body weight are subject to a prediction error corresponding to a CV of about 0.08. Since this variation is of a random nature, it is not considered in deriving the dietary energy requirements. However, in this context, where the variation in energy requirement is used for estimating the variation in energy intake, the variation owing to error in estimating the BMR is taken into account (FAO 2002). For more details on the estimation of the CV of DEC due to physiological factors refer to annex 2E.

Step 3: Aggregation Finally, the CV of DEC is derived as the sum of the square of the two CVs as estimated in steps 1 and 2.

$$CV(x) = \sqrt{(CV_{x/v}(x))^2 + (CV_{x/r}(x))^2}$$

Step 4: Selection of the CV In ADePT-FSM, the coefficient of variation of dietary energy consumption corresponds to the CV whose value is the lowest between the CV from the empirical distribution and the CV obtained as a combination of the two sources of variability.

Measures of Asymmetry

The skewness measures the asymmetry of a distribution. As opposed to income that can increase infinitely when the mean increases (corresponding to a long tail to the right), the dietary energy of food consumed is limited by biological constraints.

The method to estimate the value of the skewness depends on how the coefficient of variation of calories is derived. On one hand, when the final CV is obtained from the empirical distribution of calories across individuals, the value of the skewness is the one obtained from the empirical distribution, and a flexible distributional form is used for the calculation of the prevalence of undernourishment (PoU), known as the skewed-normal distribution. On the other hand, when the CV is derived as the sum of two components, the distribution is assumed to be lognormal and the skewness is given by:¹⁶

$$Skewness = (CV^2 + 3) * CV$$

Note that prior to the methodological improvements resulting from the choice of the CV, the addition of the skewness and the functional form of the log-normal distribution was always used for the calculation of the PoU. For this reason, the user has been left with the option of calculating the distribution according to the old or improved methodology by selecting either the log-normal or skewed-normal distribution, respectively.

Dietary Energy Requirements

The most common levels of dietary energy requirements found in the literature are the minimum and the average. They are derived from the

consideration that food energy requirements can be safely defined only in terms of a *distribution within a given class or population group*, not at the individual level (FAO/WHO/UNU 2001). Nevertheless, a minimum or average level of dietary energy intake that is compatible with a healthy and productive life can be meaningfully defined *statistically* with reference to the representative individual in a group or class. As for the estimation of both requirements, the Food and Agriculture Organization (FAO) has devised an indirect procedure based on expert recommendations on what the acceptable ranges of DER would be in groups of individuals of the same sex and age, and on the observed sex/age composition of the countries.

Minimum Dietary Energy Requirement The minimum dietary energy requirement (MDER) is estimated for each sex/age class of individuals based on the energy requirement (based on the basal metabolic rate) for the lowest acceptable body weight for that sex/age combination, adjusted for a minimal physical activity level compatible with a healthy life. Then a weighted average (the weights used are the proportions of the population in the corresponding sex/age groups) of the minimum DERs of each sex/age class is computed. Finally, the extra energy required by pregnant women is added to the weighted average to derive the minimum dietary energy requirement of a representative individual of the population.

Therefore, the information needed to estimate the MDER through the equations suggested by the joint FAO/WHO/UNU expert consultation held in 2001 is the following:

- Country birth ratio in the year of the survey (exogenous parameter)
- Structure of the population in the country by specific sex/age groups (from the survey)

Children Less Than 10 Years Old

- Body mass index (BMI) (50th percentile) (exogenous parameter)¹⁷
- Height of people in the country for specific sex/age groups (cm) (from demographic and health surveys [DHS] or literature)
- Weight gain per age (grams per day) (50th percentile) (exogenous parameter)¹⁸
- Energy per gram of weight gain (kcal) (exogenous parameter)¹⁹

- Country under-five mortality rate (U5MR) in the survey year (per 1,000 live births) (exogenous parameter)

Adolescents and Adults

- Body mass index (fifth percentile) (exogenous parameter)
- Height of people in the country for specific sex/age groups (cm) (from DHS or literature)
- The parameter used for adjusting the requirements due to the level of activity is the PAL. A PAL of 1.55 corresponds to sedentary physical activity (exogenous parameter).²⁰

The BMI is used to estimate the weight in kilograms for the attained height, while the U5MR value defines which equations should be applied to estimate the energy requirements of children less than two years old. The birth ratio is used to estimate the extra energy requirement for pregnant women. For more details refer to annex 2F.

Average Dietary Energy Requirement The formulas to estimate the average dietary energy requirement (ADER) are equal to those used in the estimation of MDER; however, some parameters are different. The ADER refers to the amount of energy considered adequate to meet the energy needs for normative average acceptable weight for an attained height while performing moderate physical activity in good health. Therefore, only the 50th percentile of the BMI is applied to all the equations. The PAL parameter to estimate the average energy requirement is 1.85 and corresponds to a moderate level of physical activity.

While no large variation is expected to exist between the metabolic rate of people in different countries within the same sex/age group (though differences across latitude could be important), the sex/age composition of the population changes over time, and so the estimated dietary energy requirements have to be adjusted to reflect this change in demographic structure.

Micronutrients Availability versus Recommended or Required Intakes

The so-called “hidden hunger” refers to a deficiency of micronutrients; it is a health threat, particularly for children and pregnant women. For an individual, the amount of micronutrients supplied in the diet should be in line

with his or her required levels of mineral and vitamins. To minimize the risk of nutrient deficit or excess, a joint FAO/WHO expert group defined the dietary requirement for a micronutrient as an intake level that meets specified criteria for adequacy. This dietary requirement is expressed in terms of an estimated average requirement (EAR) and a recommended nutrient intake (RNI). EAR is the average daily nutrient intake level that meets the needs of 50 percent of the “healthy” individuals in a particular age and sex group. RNI is the daily intake, set at the EAR plus two standard deviations, which meets the nutrient requirements of almost all apparently healthy individuals in an age and sex specific population group (FAO/WHO 2004). Therefore, to express nutrient requirements and recommended intakes for population groups, the requirements applied separately to each individual belonging to the population of analysis are summed. The individual requirements were defined for sex/age population groups by a FAO/WHO group of experts in 1998 (FAO/WHO 2004).

Despite having the micronutrient content of food acquired or consumed by households, it is not possible to talk about micronutrient consumption but availability at the household level. The reason for this is that from the moment households acquire the food to the time they eat it the content of nutrients in the food has changed. The nutrient content varies with food storage practices and processing and preparation methods (NAS 2000). For example, (1) high temperature processing can affect the vitamin content, e.g., vitamin C; and (2) discarding of water used in cooking will lead to the loss of water-soluble food components (e.g., B vitamins, vitamin C, and certain bioactive components) (FAO/INFOODS 2012).

According to the National Academy of Sciences (NAS) (2000), the household nutrient requirement estimated as the sum of the needs²¹ of the household members cannot be used as an EAR because intake and requirement are not correlated for most nutrients. When a diet provides the amount of nutrient needed by household members, it is likely that food (and so the nutrient) will be distributed in proportion to energy needs of the individuals, not to nutrient requirement needs. Therefore, it is suggested to estimate the required nutrient density of a household diet, such that when the diet is shared in proportion to calories, it is likely that the nutrient requirements of all the individuals will be met. The required nutrient density of the household is the highest nutrient density among household members (FAO/WHO 1970). Note that if the dietary energy consumed is not enough for the total household, it cannot be assured that food (and nutrient consumption)

is distributed in proportion to the calories required by household members. Therefore, this approach is meaningful when households have a consumption of calories at least equal to their requirements.

The NAS also stated that the calculation of required nutrient density is not as simple as computing the ratio of the estimated average requirement for the nutrient to the average energy requirement.²² On the one hand, the calculations must take into account variability of the nutrient requirement, expected variability of the nutrient density in ingested diets, and assurance of adequacy for the targeted individual. On the other hand, the recommended nutrient intake, which meets the nutrient requirements of almost all individuals (when requirement in the group has a normal distribution), should not be used as a cutoff point for assessing nutrient intakes of a population group because it would result in an overestimation of the proportion of people at risk of inadequacy.

Currently, ADePT-FSM generates statistics that compare levels of mean nutrient availability with both mean nutrient requirements and recommended intakes. These statistics, which are described below, will be reconsidered in light of the NAS report for future ADePT-FSM releases.

Ratio of Micronutrient Available to Required For a given population group, the average amount of available micronutrients (to be consumed by a representative individual of the population) is divided by the estimated average requirement. The available amount for consumption is the numerator, and the EAR is the denominator of the ratio. In comparing the average availability with a measure of average requirement, micronutrient distribution across the population is unaccounted.²³

Ratio of Micronutrient Available to Recommended For a given population group, the average amount of micronutrients available (to be consumed by a representative individual of the population) is divided by the recommended nutrient intake. The available amount for consumption is the numerator, and the RNI is the denominator of the ratio. For most micronutrients, if the mean intake equals the RNI, a substantial proportion of the population will have intakes less than their own requirements.

Other Ratios The development of Food-Based Dietary Guidelines (FBDG) focuses on how a combination of foods can meet nutrient requirements rather than on how each specific nutrient is provided in adequate amounts.

In contrast to recommended intakes (RI), FBDG are based on the fact that people eat food, not nutrients. Defining nutrient intakes alone is only part of the task of dealing with nutritional adequacy. The notion of nutrient density is helpful for defining FBDG and evaluating the adequacy of diets. Unlike RI, FBDG can be used to educate the public through the mass media and provide a practical guide to selecting foods by defining dietary adequacy (FAO/WHO 2004).

Food items can have a high or low nutrient density. Nutrient dense foods are those providing substantial amounts of vitamins and minerals and relatively few calories. For instance, fruits and vegetables are nutrient dense commodities. On the other hand, food items with low nutrient density supply calories but a relatively small or null amount of micronutrients. This is the case of added sugars, fats, and alcohol (USHHS/USDA 2005).

For some vitamins and minerals, ADePT-FSM estimates nutrient densities in the diet as the amount of micronutrients per 1,000 kcal provided by the food; it also estimates required/recommended nutrient densities that are estimated as the EAR/RNI of the nutrients per 1,000 required calories. The 1,000 required calories are based on the average dietary energy requirements for a representative individual of the population. Then, for these vitamins and minerals, the software computes ratios of nutrient densities in the diet to required/recommended nutrient densities.

Prevalence of Undernourishment

Since the beginning of its history FAO put emphasis on the problem of undernourishment and studied the problem in depth to reach a good estimation of the number of undernourished people in the world. The first publication on this subject was the “World Food Survey.” Starting in 1946, every five or seven years it depicted the situation of global undernourishment. Subsequently, there has been substantial improvement in the methodology used to produce this estimation and in the many experts working on that issue (especially P. Sukhatme and L. Naiken). In 1996 the first FAO World Food Summit was held, and it was decided to have a yearly publication, *The State of Food Insecurity in the World* (SOFI), which provides the latest estimates of the number of chronically hungry people in the world and introduces the first comparable estimates ever made of the number of people who go hungry. This and subsequent editions

of SOFI serve as regular progress reports on global and national efforts to reach the goal set by the World Food Summit in 1996: to reduce the number of undernourished people in the world by half by the year 2015 (FAO 1999). The term *undernourishment* indicates the condition of not consuming, on average over an extended period of time (usually a year), an amount of dietary energy sufficient to cover the minimum requirements for a healthy life.

The calculation is an exercise in model-based statistical inference. A probability distribution model is assumed for the annual average dietary energy intake of a representative individual in the population, and its parameters are estimated on the basis of the best available data. Required data include (1) the average food consumption, (2) information on the distribution of food access within the population (variability and asymmetry), (3) the demographic structure of the population (by age and sex population groups), and (4) anthropometric data. Once the probability distribution is characterized and the threshold is set, the proportion of the population that is likely suffering from chronic food deprivation, PoU, is estimated as the probability mass that falls below the threshold.

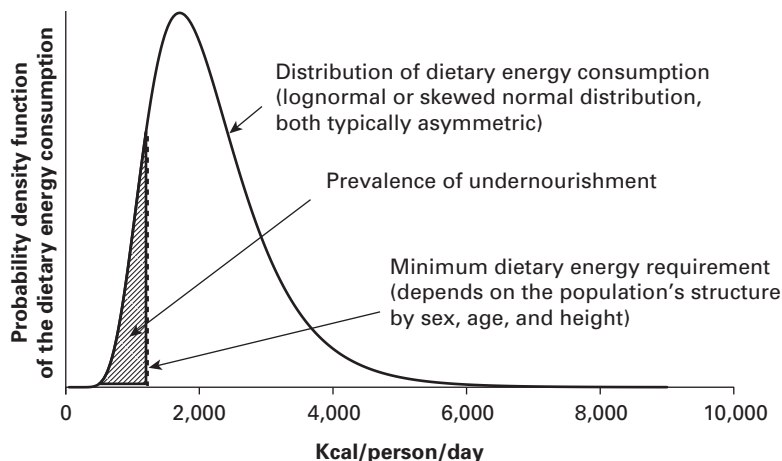
Formally, the PoU expresses the probability that, by randomly selecting one individual from the population, a person will be found to consume (on average and over a period of time) a level of food energy below the minimum required to maintain a healthy life. The operational definition of food insecurity that is embedded in this indicator is best labeled as “chronic undernourishment in a population.”

The probability distribution framework is:

$$PoU = P(x < r_L) = \int_{x < r_L} f_x(x) dx = F_x(x_{r_L})$$

where *PoU* represents the probability that an individual randomly selected within a population is found to be undernourished; *x* represents the daily habitual dietary energy consumption within a year of a “representative individual” in the population; *r_L* is the daily minimum dietary energy requirements of a “representative individual” in the population; and *f(x)* represents the distribution of yearly habitual dietary energy consumption across individuals, or, equivalently, the probability distribution of the habitual food intake levels for the population’s “representative” individual. See figure 2.2.

Figure 2.2: Graphical Representation of the Model



The major limitations of this methodology are the following:

- Though the concept relates to an individual condition, the indicator is designed to measure hunger at the level of a population. Its calculation neither depends on the possibility of collecting data on individuals, nor is the indicator intended to be used to assess the undernourishment condition of any specific individual or group of individuals in a reference population. Therefore, it does not capture possible idiosyncratic, individual problems in accessing food. However, the method can be applied to subnational populations, provided subgroups are representative of the population and data pertaining to such subpopulations are available (Sibrián 2008).
- By focusing on food access, it does not reflect cases of *malnutrition* associated with factors related to the efficient utilization of food.
- Similarly, it misses the “quality” dimension of food security, for example, micronutrient deficiency and related morbidity.
- Finally, by focusing on a determined period, the indicator misses the dimension of risk and vulnerability associated with the (in)stability in the access to food.

Implementing the statistical concept just described to NHS data requires a set of *ancillary assumptions*, mostly driven by feasibility and

data availability constraints. The current practice at FAO is based on the following:

1. Food intake is approximated by quantities *available* for consumption at the household level, with no consideration of household level food waste.
2. The distribution of food available for consumption is analyzed between households. Therefore, possible unequal distribution of food within the household is ignored.
3. The minimum dietary energy requirement is defined at the population and not at the individual level.
4. An assumption needs to be made regarding the distribution of the usual intake of a typical person from the population.

The first two assumptions are conditioned by data availability. Though measures of actual intake could be obtained from nutrition surveys, the vast majority of available datasets on food consumption do not allow for a precise estimation of household food waste. Similarly, data from very few and recent surveys could allow for an analysis of the intrahousehold distribution of food consumption.

Assumption 3 is more substantial. It is derived from the consideration that food energy requirements can be safely defined only in terms of a *distribution within a given class or population group*, not at the individual level (FAO/WHO/UNU 2001). This implies that classification of single individuals as undernourished based on a comparison of the level of habitual food intake with their individual requirements is problematic, because the latter cannot usually be estimated with sufficient precision. A minimum level of dietary energy intake that is compatible with a healthy and productive life can nevertheless be meaningfully defined in a *statistical sense* with reference to the representative individual in a group or class.

Assumption 4 regards the statistical model used to conduct the inference at the population level. Originally, the log-normal distribution was chosen due to some desirable characteristics. For example, it is positive valued and with an elongated right tail, and the number of parameters needed for its characterization (only two: one for location and one for dispersion). Concerns were raised that the log-normal model may be not flexible enough to capture changes in the distribution of food access, especially if such changes affect the two “tails” of the distribution in opposite ways. For this

reason, in 2011, FAO's Statistics Division explored alternative models that afford greater flexibility in representing the distribution of food consumption.

Therefore, a more flexible model (the skew-normal introduced by Azzalini in 1980) has been deemed more appropriate to represent the distribution of habitual food consumption in the population; a major advantage compared to the log-normal distribution is that this model can now capture changes in the asymmetry of the distribution of food consumption.

Estimation of PoU in ADePT

ADePT-FSM provides estimates of undernourishment (derived from NHS data) at the national level, for urban and rural areas, and regions provided that these population groups have representativeness in the survey sample.

MDG 1.9 Indicator of Prevalence of Food Deprivation

In addition to the estimation of the PoU using NHS data, with ADePT-FSM it is possible to estimate, at the year of the survey, the value of the prevalence of undernourishment used to compute the MDG 1.9 indicator published yearly by FAO in SOFI. Indeed, one of FAO's mandates is to provide a global estimate of the prevalence of undernourishment to monitor progress toward reduction of global hunger by half by 2015 compared to the level of 1990–92. To publish estimates of the prevalence of undernourishment for about 180 countries back to 1990–92, FAO is using the dietary energy supply (DES) as derived from the food balance sheets and corrected for losses at the retail level as the mean of the distribution.

The MDG 1.9 indicator is computed as a three-year weighted average of the number of people undernourished using total population as weights. The PoU for a specific year used in the three-year average can be reproduced using ADePT-FSM if data²⁴ on the DES, adjusted by losses at the retail level, CV, skewness, and MDER, are used as exogenous parameters in ADePT.

Besides the MDG 1.9 indicator, ADePT-FSM also computes the PoU by region, urban, and rural areas. In these cases the values of CV and skewness remain the same while the DES adjusted by losses, and MDER are calculated by the program applying the formulas:

$$DES_{Sub} = DES_{Nat} * \frac{DEC_{SubHS}}{DEC_{NatHS}}$$

$$MDER_{Sub} = MDER_{SOFI} * \frac{MDER_{SubHS}}{MDER_{NatHS}}$$

where DES_{Sub} is the individual daily dietary energy supply (corrected for losses²⁵) at the regional or area level; DES_{Nat} is the individual daily dietary energy supply (corrected for losses²⁶) at the national level; DEC_{NatHS} is the daily dietary energy consumption per person at the national level obtained from the NHS data; DEC_{SubHS} is the daily dietary energy consumption per person at the regional or area level obtained from the NHS data; $MDER_{Sub}$ is the daily minimum dietary energy requirement per person at the regional or area level; $MDER_{SOFI}$ is the daily minimum dietary energy requirement per person at the national level as used in SOFI; $MDER_{NatHS}$ is the daily minimum dietary energy requirement per person at the national level obtained from the NHS data; and $MDER_{SubHS}$ is the daily minimum dietary energy requirement per person at the regional or area level obtained from the NHS data.

Depth of the Food Deficit

The depth of the food deficit indicates how many calories would be needed to lift the undernourished from their status, everything else being constant. The average intensity of food deprivation of the undernourished, estimated as the difference between the average dietary energy requirement and the average dietary energy consumption of the undernourished population, is multiplied by the number of undernourished to provide an estimate of the total food deficit in the country, which is then normalized by the total population.

For each category of the population groups, the depth of the food deficit has to be estimated as the absolute difference between the average calories consumed by the deprived population and the average dietary energy requirements multiplied by the prevalence of undernourishment.

The average consumption of the undernourished population can be computed by taking the average consumption corresponding to the part of the distribution of dietary energy consumption below the minimum dietary energy requirement, as follows:

$$\mu_u = \frac{\int_0^{MDER} xf(x) dx}{\int_0^{MDER} f(x) dx}$$

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where μ_u is the average dietary energy consumption of the food-deprived population; $f(x)$ is the density function of dietary energy consumption; and $MDER$ is the minimum dietary energy requirement.

Absolute food deficit from average dietary energy requirements (ADER) in food-deprived population

$$\text{Food Deficit} \left(\frac{\text{Kcal}}{\frac{\text{person}}{\text{day}}} \right) = ADER - \mu_u$$

Depth of food deficit is then estimated as:

$$\text{Depth of food deficit} \left(\frac{\text{Kcal}}{\frac{\text{person}}{\text{day}}} \right) = (ADER - \mu_u) * PoU$$

where PoU refers to the prevalence of undernourishment.

Annexes

Annex 2A

Table 2A.1: Example of Different Units of Measurement in Which Food Data Are Collected and Respective Conversion into Metric Units

<i>Food item in survey</i>	<i>Unit of measurement of the food quantity in survey</i>	<i>Unit equivalent to convert unit of measurement into (A)</i>	<i>Food item quantity collected in survey (B)</i>	<i>Density coefficient (gram/milliliter) (C)</i>	<i>Extra factor (D)</i>	<i>Food item quantity standardized in (E = A*B*C*D)</i>
<i>Kilogram or liter</i>			<i>Kilogram</i>			
Flour of wheat	Kilogram	1	2.5	1	1	2.5
Flour of wheat	Gram	0.001	500	1	1	0.5
Beer	Liter	1	3	1.007	1	3
Beer	Milliliter	0.001	1,500	1.007	1	1.5
Egg	Unit of 50 grams	0.05	12	1	1	0.6
Rice	Bag of 5 kilograms	5	1	1	1	5
Crackers	Packet of 500 grams	0.5	2	1	1	1
Tuna	Pound	0.45359	0.5	1	1	0.23
<i>Gram or milliliter</i>			<i>Gram</i>			
Flour of wheat	Kilogram	1,000	2.5	1	1	2,500
Flour of wheat	Gram	1	500	1	1	500
Beer	Liter	1,000	3	1.007	1	3,021
Beer	Milliliter	1	1,500	1.007	1	1,500
Egg	Unit of 50 grams	50	12	1	1	600
Rice	Bag of 5 kilograms	5,000	1	1	1	5,000
Crackers	Packet of 500 grams	500	2	1	1	1,000
Tuna	Pound	453.59	0.5	1	1	226.8
<i>Gram or milliliter</i>			<i>Pound</i>			
Flour of wheat	Kilogram	1,000	2.5	1	0.0022046	5.51
Flour of wheat	Gram	1	500	1	0.0022046	1.1
Beer	Liter	1,000	3	1.007	0.0022046	6.66
Beer	Milliliter	1	1,500	1.007	0.0022046	3.31
Egg	Unit of 50 grams	50	12	1	0.0022046	1.32
Rice	Bag of 5 kilograms	5,000	1	1	0.0022046	11.02
Crackers	Packet of 500 grams	500	2	1	0.0022046	2.2
Tuna	Pound	453.59	0.5	1	0.0022046	0.5

Table 2B.2: Procedure 1: Steps 5 to 6

HH ID	From food items having food quantities expressed in standard units and valid nutrient conversion factors (g/person/day)					Calories (kcal/person/day) from the following:					Calories from food items having valid quantities and nutrient conversion factors (kcal/person/day)				
	Available					Available					Kcal = G + H + I + J + K				
	Protein	Fats	carbohydrates	Fiber	Alcohol	Protein	Fats	carbohydrates	Fiber	Alcohol					
	B	C	D	E	F	B*4	C*9	D*4	E*2	F*7					
						G	H	I	J	K	L				
1	74	63	530	28	0	296	566	2,119	57	0	3,038				
2	28	29	304	51	0	110	258	1,215	103	0	1,686				
3	72	83	646	100	0	288	751	2,584	199	0	3,822				
4	29	45	332	38	0	115	403	1,328	76	0	1,922				
5	15	52	157	22	0	60	466	629	43	0	1,199				
6	19	50	245	17	0	77	447	981	34	0	1,539				
7	68	58	274	38	0	270	518	1,095	75	0	1,959				
8	35	62	479	42	0	138	560	1,917	84	0	2,699				
9	41	37	203	38	0	164	329	813	76	0	1,382				
10	67	103	448	25	0	267	924	1,790	51	0	3,032				
11	33	84	248	44	0	134	757	993	87	0	1,971				
12	42	62	354	28	0	168	556	1,415	56	0	2,194				
13	21	39	329	72	0	83	349	1,317	144	0	1,893				
14	31	57	462	32	0	124	509	1,848	63	0	2,543				
15	30	87	537	31	0	122	785	2,148	62	0	3,117				
16	39	57	534	56	0	154	515	2,137	111	0	2,918				
17	44	49	276	21	0	176	438	1,104	42	0	1,759				
18	35	46	277	84	0	140	411	1,106	168	0	1,825				
19	25	57	381	34	0	99	513	1,522	67	0	2,202				

Annex 2C

Procedure 2: Estimation of Nutrients and Calories from Food Expenditure

Table 2C.1: Procedure 2: Steps 1 to 2

Food items having food quantities expressed in standard units and valid nutrient conversion factors												
HH ID	Calories (kcal/ person/ day)	Proportion of calories from		Food expenditure (lcu/person/ day)	Household calorie price (lc/kcal)	Region	Area	Income quintile	Average calorie price (\$/kcal) of Region 1, urban area and income quintile 2	Food expenditure associated to missing calories (lcu/person/day)	Estimated missing calories (Kcal/ person/day)	O
		protein 100*G/L	fats 100*H/L									
1	3,038	10	19	2.4	0.0008	1	Urban	2	0.00083	1.8	2,147	
2	1,686	7	15	1.2	0.0007	1	Urban	2	0.00083	1.3	1,604	
3	4,947	8	20	3.3	0.0009	1	Urban	2	0.00083	1.2	1,448	
4	1,922	6	21	1.2	0.0006	1	Urban	2	0.00083	0.8	1,001	
5	1,199	5	39	1.2	0.0010	1	Urban	2	0.00083	1.1	1,339	
6	1,539	5	29	0.7	0.0005	1	Urban	2	0.00083	0.9	1,122	
7	1,959	14	26	2.6	0.0013	1	Urban	2	0.00083	0.4	531	
8	2,699	5	21	6.6	0.0024	1	Urban	2	0.00083	0.9	1,074	
9	1,382	12	24	2.3	0.0016	1	Urban	2	0.00083	0.5	639	
10	3,032	9	30	2.1	0.0007	1	Urban	2	0.00083	0.7	808	
11	1,971	7	38	1.4	0.0007	1	Urban	2	0.00083	1.3	1,604	
12	2,194	8	25	1.4	0.0006	1	Urban	2	0.00083	0.7	808	
13	1,893	4	18	1.4	0.0008	1	Urban	2	0.00083	0.5	639	
14	2,543	5	20	1.2	0.0005	1	Urban	2	0.00083	0.6	663	
15	3,117	4	25	1.6	0.0005	1	Urban	2	0.00083	0.5	639	
16	2,918	5	18	1.5	0.0005	1	Urban	2	0.00083	0.4	434	
17	1,759	10	25	1.7	0.0010	1	Urban	2	0.00083	0.5	627	
18	1,825	8	23	1.6	0.0009	1	Urban	2	0.00083	0.5	603	
19	2,202	4	23	1.0	0.0005	1	Urban	2	0.00083	0.5	603	

Table 2C.2: Procedure 2: Steps 3 to 5

HH ID	Estimated missing calories (kcal/person/day) from		Estimated missing calories (kcal/person/day) from carbohydrates (including fiber and alcohol)	Estimated missing grams (person/day)		
	Protein	Fats		Protein	Fats	Carbohydrates (including fiber and alcohol)
	P	Q		S	T	U
	$O*M/100$	$O*N/100$		$P/4$	$Q/9$	$R/4$
1	209	400	1,538	52	44	384
2	105	246	1,254	26	27	313
3	109	284	1,054	27	32	264
4	60	210	731	15	23	183
5	67	521	751	17	58	188
6	56	326	740	14	36	185
7	73	140	317	18	16	79
8	55	223	796	14	25	199
9	76	152	411	19	17	103
10	71	246	491	18	27	123
11	109	617	879	27	69	220
12	62	205	542	15	23	135
13	28	118	494	7	13	123
14	32	133	499	8	15	125
15	25	161	453	6	18	113
16	23	77	335	6	9	84
17	63	156	409	16	17	102
18	46	136	421	12	15	105
19	27	141	435	7	16	109

Annex 2D

Table 2D.1: Example of Calculation of Food and Total Price Temporal Deflators

Month	Year	Price indexes		Deflators	
		Food price index (FPI)	Consumer price index (CPI)	Food monetary values deflator (FPI/average FPI)	Total consumption and income deflator (CPI/average CPI)
9	2004	121.91	122.44	0.962	0.958
10	2004	123.04	123.80	0.971	0.968
11	2004	123.74	124.73	0.977	0.976
12	2004	123.85	124.62	0.978	0.975
1	2005	124.61	125.30	0.984	0.980
2	2005	125.54	126.35	0.991	0.988
3	2005	126.61	127.69	0.999	0.999
4	2005	127.77	129.14	1.009	1.010
5	2005	129.79	131.33	1.024	1.027
6	2005	131.75	134.01	1.040	1.048
7	2005	131.56	133.66	1.038	1.046
8	2005	130.10	131.02	1.027	1.025
Average		126.69	127.84		

Annex 2E

Table 2E.1: Estimation of the Coefficient of Variation of Dietary Energy Consumption Due to Other Factors

$CV^2_{x/r} = CV^2_{PAL} + CV^2_{wh} + CV^2_{err}$		
Where Ln refers to Neperian Logarithm, and normsinv(p) is the function which returns the value Z such that, with probability p, a standard normal random variable takes on a value that is less than or equal to Z.		
(CV_{PAL}) Coefficient of variation of dietary energy requirement due to physical activity level (PAL)		
Standard deviation of PAL – Maximum (MXSDP)	$(Ln(2.4)-Ln(1.4))/(normsinv(0.95) - normsinv(0.05))$	
Standard deviation of PAL – Minimum (MNSDP)	$(Ln(2.4)-Ln(1.4))/(normsinv(0.975) - normsinv(0.025))$	
CV of PAL – Maximum	$= \sqrt{(e^{MXSDP^2})} - 1$	
CV of PAL – Minimum	$= \sqrt{(e^{MNSDP^2})} - 1$	
<i>If the proportion of labor force in the primary sector is more than 49%</i>		
CV of dietary energy requirement due to PAL	Maximum CV of PAL * proportion of labor force in the primary sector	
<i>If the proportion of labor force in the primary sector is less than 50%</i>		
CV of dietary energy requirement due to PAL	Minimum CV of PAL * proportion of labor force in the primary sector	
(CV_{wh}) Coefficient of variation of dietary energy requirement due to body weight		
For each sex/age population group	Weight in Kg using the 5th percentile of the BMI and the median height in cm (W5th)	$BMI_{5th} * \left(\frac{height}{100}\right)^2$
	Weight in Kg using the 95th percentile of the BMI and the median height in cm (W95th)	$BMI_{95th} * \left(\frac{height}{100}\right)^2$
	Variance of the distribution of body weight	$(Ln(W95th)-Ln(W5th))/(normsinv(0.95)- normsinv(0.05))^2$
Population weighted average value of the variance (VARBW)	$\sum_{i=1}^{31} (f_i * \delta_i^2)$	Where <i>i</i> refers to the sex/age group; <i>f</i> is the proportion of the population belonging to the <i>ith</i> group; and <i>δ</i> is the standard deviation of the distribution of body weight of the <i>ith</i> group.
CV of dietary energy requirement due to body weight	$= \sqrt{(e^{VARBW})} - 1$	
(CV_{err}) Coefficient of variation due to error		
CV of dietary energy requirement due to error	0.08	

Annex 2F

Table 2F.1: Estimation of the Minimum Dietary Energy Requirement

Where TEE refers to total energy expenditure (kcal); U5MR refers to under-five mortality rate; the probability per 1,000 that a newborn baby will die before reaching age 5, if subject to current age-specific mortality rates; KG refers to BMI * (height/100)^2; height is in cm; WG refers to weight gain for age (g/day); ERwg is the energy required per gram of weight gain (kcal); and PAL refers to 1.55 for sedentary physical activity.			
Years: Less than 1/Class group: 1			Note
Male and female	Country with high children undernutrition and infection (U5MR proxy high)		50th percentile for BMI and WG
	if U5MR > 10‰	TEE = (−99.4 + 88.6 * KG) + 2 * WG * ERwg	
	Country with low children undernutrition and infection (U5MR low)		
	if U5MR <= 10‰	TEE = (−99.4 + 88.6 * KG) + WG * ERwg	
Years: From 1 to 1.9/Class group: 2			Note
	Country with high children undernutrition and infection (U5MR proxy high)		50th percentile for BMI and WG
Male	if U5MR > 10‰	TEE = 0.93 * (310.2 + 63.3 * KG − 0.263 * KG²) + 2 * WG * ERwg	
Female	if U5MR > 10‰	TEE = 0.93 * (263.4 + 65.3 * KG − 0.454 * KG²) + 2 * WG * ERwg	
Country with low children undernutrition and infection (U5MR low)			
Male	if U5MR <= 10‰	TEE = 0.93 * (310.2 + 63.3 * KG − 0.263 * KG²) + WG * ERwg	
Female	if U5MR <= 10‰	TEE = 0.93 * (263.4 + 65.3 * KG − 0.454 * KG²) + WG * ERwg	
Years: From 2 to 9.9/Class group: From 3 to 10			
Male	TEE = (310.2 + 63.3 * KG − 0.263 * KG²) + WG * ERwg		50th percentile for BMI and WG
Female	TEE = (263.4 + 65.3 * KG − 0.454 * KG²) + WG * ERwg		
Years: From 10 to 17.9/Class group: From 11 to 18			Note
Male	TEE = 0.85 * (310.2 + 63.3 KG − 0.263 KG²) + WG * ERwg		5th percentile for BMI and WG
Female	TEE = 0.85 * (263.4 + 65.3 KG − 0.454 KG²) + WG * ERwg		
Years: From 18 to 29.9/Class group: From 19 to 22			Note
Male	TEE = PAL * (692.2 + 15.057 KG)		5th percentile for BMI
Female	TEE = PAL * (486.6 + 14.818 KG)		
Years: From 30 to 59.9/Class group: From 23 to 28			Note
Male	TEE = PAL * (873.1 + 11.472 KG)		5th percentile for BMI
Female	TEE = PAL * (845.6 + 8.126 KG)		
Years: More than 59.9/Class group: From 29 to 31			Note
Male	TEE = PAL * (587.7 + 11.711 KG)		5th percentile for BMI
Female	TEE = PAL * (658.5 + 9.082 KG)		
Pregnancy allowance			
Energy extra = Birth ratio * 210			

Notes

1. Expressed as per person per day.
2. Further information in forthcoming *Assessment of the Reliability and Relevance of the Food Data Collected in National Household Consumption and Expenditure Surveys*.
3. Note that tobacco and narcotics are not considered as food and are excluded from the analysis.
4. For more information on food matching, see the FAO/INFOODS *Guidelines for Food Matching, Version 1.2* (2012). The *Guidelines for Food Matching* include a quality scheme for the food matches. They should be recorded and used in the assessment of the quality of the estimated energy and nutrient intake and food security estimations. Food composition data are not just values that can be used without previous knowledge on food composition. If done so, there is a high risk of applying the data wrongly. Therefore, it is highly recommended to complete the FAO/INFOODS e-Learning Course on Food Composition Data (available free-of-charge at the INFOODS website).
5. For converting amounts of total foods (including inedible part) to EP, see the FAO/INFOODS *Guidelines for Converting Units, Denominators and Expressions Version 1.0* (2012).
6. Density values for liquids or semiliquids can be found in the FAO/INFOODS Density Database Version 2.0: <http://www.fao.org/infoods/infoods/tables-and-databases/faoinfoods-databases/en/>.
7. If the data were corrected for outliers the median should be close to the mean.
8. Density values of solids to be applied in this analysis are equal to 1.
9. *Guidelines for Converting Units, Denominators and Expressions* is available at <http://www.fao.org/infoods/infoods/standards-guidelines/en/>.
10. Unless it is a nutritional dietary survey that measures direct individual food intake.
11. For instance, ascorbic acid enhances iron absorption.
12. When estimating the available micronutrients for consumption, ADePT-FSM excludes from the analysis those provided by food consumed away from home.
13. Heme iron accounts for a minor part of total iron intake, especially in developing countries where the consumption of meat and fish is usually low. Thus, nonheme iron is the main source of dietary iron intake

(Hallberg 1981). Still, not all the nonheme iron consumed is absorbed by the human body because this process is influenced either positively or negatively by many factors, such as the presence of certain substances in the diet.

14. The indispensable amino acids are not present in all food items. For instance, lysine, threonine, and tryptophan are marginal in cereals. Whereas the former (lysine) is lacking especially in wheat, the latter (tryptophan) is lacking in maize. Methionine and cysteine, which are equally abundant in cereal and animal proteins, are marginal in legume proteins (WHO 2007).
15. When estimating the available amino acids for consumption, ADePT-FSM excludes from the analysis those provided by food consumed away from home.
16. As the log-normal distribution is fully characterized by only two parameters (μ and σ), the skewness coefficient is a simple monotonic transformation of the standard deviation $SK = (e^{\sigma^2} + 2)\sqrt{e^{\sigma^2} - 1}$, and it can also be conveniently expressed as a function of the coefficient of variation, according to the formula
$$\text{skewness} = (CV^2 + 3) \times CV.$$
17. See WHO Child Growth Standards: BMI for age tables. WHO. Geneva. http://www.who.int/childgrowth/standards/bmi_for_age/en/index.html. In addition, see WHO 2007, WHO Growth reference 5–19 years: BMI for age tables. WHO. Geneva. http://www.who.int/growthref/who2007_bmi_for_age/en/index.html.
18. For further information see WHO (1983).
19. For further information see FAO/WHO/UNU (2001).
20. For further information see FAO/WHO/UNU (2001).
21. The sum of the needs is determined by using the average of needs of similar individuals.
22. Based on a publication FAO/WHO/UNU (1985).
23. The prevalence of inadequacy depends on the shape and variation of the usual intake distribution, not on mean intake. See NAS (2000).
24. See the FAO statistics website: <http://www.fao.org/economic/ess/ess-fs/fs-methods/adept-fsn/en/>.
25. The losses are at the retail level and exclude losses within households.
26. Based on a publication FAO/WHO/UNU (1985).

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